PROFESSIONAL EDUCATION



SUMMER 2023

GRAPH ALGORITHMS AND MACHINE LEARNING

DATES	DURATION	FORMAT	FEE	CEUS
July 31–August 3, 2023	2 Days	Live Virtual	$2,500	1.4

Enhance your ability to extract useful insights from large and structured data sets

The efficient processing of large graphs has attracted significant attention, due to its applications in various domains, including social network analysis, epidemiology, computational biology, machine learning, and scientific simulations. Today, graphs have become extremely large and are evolving rapidly in real-time—which has made designing graph analytics a major challenge.

This accelerated course provides a comprehensive overview of critical topics in graph analytics, including applications of graphs, the structure of real-world graphs, fast graph algorithms, synthetic graph generation, performance optimizations, programming frameworks, and learning on graphs. The curriculum additionally covers software performance engineering concepts, such as parallelism, caching, and compression, in the context of graph processing, as well as different design choices that will enable you to use or design the appropriate graph solutions for your needs.

Through tutorials, exercises, and demonstrations featuring state-of-the-art graph analytics tools, you will broaden your fundamental understanding of graph analytics, and master the techniques and tools that you need to efficiently solve large-scale graph problems in your organization.

WHO SHOULD ATTEND

This course is designed for scientists and engineers in industry or government who work with large-scale data. The strategies covered are applicable to a variety of fields, such as software/IT, finance, transportation, biotech, telecommunications, and cybersecurity.

Applicable roles include, but are not limited to:

- **Data scientists** who want to improve their ability to extract actionable insights from large and structured data sets
- **Software engineers** looking to develop fast graph software
- **Project managers** who wish to increase their efficiency in overseeing technical teams working on graph-related projects and graph software products
- **Any technical professionals** who want to use data to inform business decisions, accelerate discoveries, detect fraud, defend against security threats, increase revenue, or improve service quality

EARN A PROFESSIONAL CERTIFICATE

This course may be taken individually or as part of the ***Professional Certificate Program in Machine Learning & Artificial Intelligence.***

professional.mit.edu/gaml

LEARNING OUTCOMES

By participating in this course, you will:

- Learn how to model structured data with graphs
- Enhance your understanding of real-world graph properties and how to generate synthetic graphs
- Master fundamental graph algorithms
- Discuss parallelism and how it can be used to speed up graph processing
- Examine performance characteristics of graph algorithms
- Assess the state-of-the-art graph processing tools available today and learn to use certain graph software
- Explore the pros and cons of different graph processing approaches
- Acquire a new set of tools for improving the effectiveness and performance of machine learning pipelines

LEAD INSTRUCTOR

Julian Shun



Julian Shun is an Associate Professor of Electrical Engineering and Computer Science at MIT and a lead investigator in MIT Computer Science and Artificial Intelligence Laboratory (CSAIL). His research focuses on the theory and practice of parallel algorithms and programming, with particular emphasis on designing algorithms and frameworks for large-scale graph processing and spatial data analysis. He also studies parallel algorithms for text analytics, concurrent data structures, and methods for deterministic parallelism. Prior to joining MIT, he was a postdoctoral Miller Research Fellow at UC Berkeley. His honors include the NSF CAREER award, DOE Early Career Award, ACM Doctoral Dissertation Award, the CMU School of Computer Science Doctoral Dissertation Award, Google Faculty Research Award, Google Research Scholar Award, SoE Ruth and Joel Spira Award for Excellence in Teaching, Facebook Graduate Fellowship, and best paper awards at PLDI, SPAA, CGO, and DCC.

Teaching Methodology

The certificate curriculum is grounded in the spirit of MIT's motto, "Mens et Manus," or "mind and hand," which combines theoretical instruction with hands-on methods of discovery.



INTERACTIVE LECTURES



HANDS-ON EXERCISES



GROUP PROJECTS



SUPPLEMENTAL RESOURCES

PROFESSIONAL EDUCATION

ABOUT MIT PROFESSIONAL EDUCATION

For 70 years, MIT Professional Education has been providing technical professionals worldwide a gateway to renowned MIT research, knowledge, and expertise, through advanced education programs designed specifically for them. In addition to industry-focused, two-to-five-day live virtual and on-campus courses through Short Programs, MIT Professional Education offers professionals the opportunity to take online and blended learning courses through Digital Plus Programs, attend courses abroad through International Programs, enroll in regular MIT academic courses through the Advanced Study Program, or attend Custom Programs designed specifically for their companies. For more information, please visit **professional.mit.edu**.